

Interval-censored Hawkes Processes via the Mean Behavior Poisson Process

From Poisson and Hawkes to covariate-modulated excitation,
and the Volterra systems that arise

A self-contained mathematical treatment
companion to the `hawkes_calibration` package

Abstract

The Hawkes process is the standard model for self-exciting events, but its likelihood requires the exact event times. In many applications we observe only *interval-censored* data: counts of events per time bucket. This document develops, from first principles and with full rigour, the machinery that makes Hawkes calibration possible in that setting — the *Mean Behavior Poisson process* (MBPP) — and then pushes it through three levels of model complexity: the bare process, time-varying *baseline* covariates, and finally time-varying *excitation* covariates. The last is the hard case: it turns the governing equation from a convolution into a *general Volterra integral equation of the second kind*, costing us the Laplace/Fourier shortcuts but still solvable. Each result states its precise hypotheses and is accompanied by an intuition. Appendices give self-contained tutorials on Volterra equations, the Laplace and Fourier transforms, and ODE/state-space methods — everything the main text relies on.

Contents

1	Introduction: the problem, and how we solve it	2
1.1	The problem	2
1.2	How we solve it	3
1.3	Roadmap, in increasing order of difficulty	3
2	Poisson and Hawkes	4
2.1	Counting processes, intensity, compensator	4
2.2	The Poisson process and why deterministic Λ matters	4
2.3	The likelihood of a point process	5
2.4	The Hawkes process and why it is <i>not</i> Poisson	5
2.5	The cluster representation and the mean-intensity equation	5
3	Standing assumptions for our model	6
4	The MBPP without covariates	7
4.1	Existence, uniqueness, and the resolvent	7
4.2	The exponential kernel closes everything	7
4.3	Multivariate MBPP	8
4.4	Fitting on interval-censored counts (IC-LL)	8

5	Adding covariates to the baseline	9
5.1	Why this stays easy	9
5.2	Fitting the covariate effects	9
5.3	What is and is not identified	9
6	Excitation covariates: the Volterra system that arises	10
6.1	The model	10
6.2	The mean-intensity equation is now a general Volterra equation	10
6.3	What survives, and what is lost	10
6.4	Exponential shape $\Rightarrow$ a finite linear time-varying ODE	11
6.5	Fitting with excitation covariates	11
6.6	Summary of the three regimes	12
A	Volterra integral equations: a primer	12
A.1	What is an integral equation, and why “Volterra”?	12
A.2	The operator form and the Neumann (resolvent) series	13
A.3	Existence and uniqueness: why Volterra is so forgiving	13
A.4	Three solution strategies, by kernel type	14
B	Laplace and Fourier methods for convolution kernels	14
B.1	The transforms and the convolution theorem	15
B.2	Solving the convolution Volterra equation	15
B.3	The spectral (neural-operator) viewpoint, and why it is exact here	15
B.4	Why transforms fail under excitation covariates	15
C	ODE and state-space methods	16
C.1	The exponential kernel gives a first-order ODE	16
C.2	Rational Laplace transform $\Leftrightarrow$ finite ODE	16
C.3	Sums of exponentials: the state-space realisation	16
C.4	Numerical integration and the time-varying case	17
D	Estimation, Bregman geometry, and identifiability	17
D.1	IC-LL is a Bregman divergence	17
D.2	Consistency, information, and goodness of fit	17
D.3	The κ - θ ridge, and the cost of covariates	18

1 Introduction: the problem, and how we solve it

1.1 The problem

A *point process* records the times $t_1 < t_2 < \dots$ of events. The *Hawkes process* is the canonical model when events beget events — an earthquake makes aftershocks more likely, a tweet makes re-tweets more likely. Its defining feature is a *self-exciting* conditional intensity

$$\lambda(t) = s(t) + \sum_{t_i < t} \phi(t - t_i), \quad (1)$$

where $s \geq 0$ is the exogenous “background” rate that injects *immigrant* events and the kernel $\phi \geq 0$ describes how each past event lifts the future rate, producing *offspring*. To fit (1) by maximum likelihood one needs every event time t_i .

In practice we frequently *do not* observe event times. We observe *interval-censored* data: the *number* of events in each time bucket — hospital admissions per day, video views per day, transactions per hour — but not when each event occurred, whether for privacy or because that is simply how the data are recorded. Two facts then block a direct fit:

1. the log-likelihood of (1) is *undefined* without the times t_i ; and
2. the Hawkes process lacks the *independent-increments* property, so we cannot fall back on a Poisson likelihood for the bucket counts either (its intensity, hence the number of events in a bucket, depends on the counts in earlier buckets).

1.2 How we solve it

The escape is to fit a *different but parameter-matched* process whose mean behaviour mimics the Hawkes process while enjoying independent increments. This is the *Mean Behavior Poisson process* (MBPP): the inhomogeneous Poisson process whose deterministic intensity equals the *expected* Hawkes intensity,

$$\xi(t) := \mathbb{E}[\lambda(t)] = s(t) + \int_0^t \phi(t-u) \xi(u) \, du. \quad (2)$$

Three observations make this work, and they organise the whole document.

- **Why the MBPP is Poisson.** A simple point process whose compensator is *deterministic* is automatically Poisson (Watanabe’s theorem, §2); replacing the random Hawkes intensity λ by its mean ξ removes the feedback and manufactures independent increments. The bucket counts of the MBPP are therefore independent Poisson variables with means $\Xi(o_{i-1}, o_i] = \int_{o_{i-1}}^{o_i} \xi$, and their negative log-likelihood — the *interval-censored log-likelihood* — is a tractable sum of Poisson terms.
- **Why we can compute ξ .** Equation (2) is a *Volterra integral equation of the second kind*. Because ϕ here depends only on the elapsed time $t - u$, it is a *convolution*, so the system is linear and time-invariant and we can solve it in closed form (an impulse response), in the Laplace/Fourier domain (a transfer function), or as a linear ODE. Appendix A is a self-contained primer on Volterra equations; Appendices B and C cover the transform and ODE methods.
- **Why the parameters transfer.** The MBPP and the Hawkes process are built from the *same* s and ϕ , so fitting the MBPP on the counts recovers (approximations of) the Hawkes parameters.

1.3 Roadmap, in increasing order of difficulty

We deliberately build up the model one layer at a time.

§2 Poisson and Hawkes.

The two ingredients: the Poisson process (and exactly why a deterministic compensator makes a process Poisson) and the Hawkes process (and exactly why it is *not* Poisson), with the cluster/branching picture that yields (2) rigorously.

§3 Standing assumptions.

The precise regularity conditions under which everything below holds, stated once.

§4 Without covariates.

The univariate and multivariate MBPP. Here ϕ is a fixed convolution kernel, the equation is LTI, and everything is closed-form. This is the workhorse.

§5 Baseline covariates.

Let the immigrant rate depend on covariates, $s(t) = \exp(\gamma_0 + \gamma^\top X(t))$. The equation stays a convolution (only the forcing changes), so the LTI machinery survives unchanged; we simply fit the coefficients γ .

§6 Excitation covariates (the hard case).

Let the *excitation itself* depend on covariates, $\alpha(t) = \alpha_0 \exp(\delta^\top Z(t))$. Now the kernel depends on t and u *separately*, the equation is a *general (non-convolution) Volterra equation*, and time-invariance — together with the Laplace/Fourier shortcuts — is lost. We show the problem is still well-posed and, for exponential kernels, reduces to a *linear time-varying* ODE that we can integrate.

The short answer to “can we put covariates in the excitation?” is *yes* — §6 does it — but it is genuinely harder, and seeing exactly *why* is the reason the Volterra viewpoint is worth developing carefully.

2 Poisson and Hawkes

We need two building blocks: the Poisson process (which the MBPP will be) and the Hawkes process (which we want to fit). The crucial contrast is whether the *compensator* is deterministic.

2.1 Counting processes, intensity, compensator

A simple temporal point process is described by its counting process $N(t) = \#\{i : t_i \leq t\}$, adapted to its internal history $\mathcal{H}_t = \sigma\{N(s) : s \leq t\}$. By the Doob–Meyer decomposition there is a unique predictable increasing *compensator* Λ with

$$M(t) = N(t) - \Lambda(t) \text{ an } (\mathcal{H}_t)\text{-local martingale.} \quad (3)$$

When Λ is absolutely continuous, $\Lambda(t) = \int_0^t \lambda$, and the density λ is the *conditional intensity* $\lambda(t) = \lim_{h \downarrow 0} \frac{1}{h} \mathbb{E}[N(t+h) - N(t) \mid \mathcal{H}_t]$.

Intuition. Λ is the predictable “running forecast” of how many events should have happened by time t ; λ is that forecast’s instantaneous rate. Subtracting the forecast from reality leaves pure surprise — the martingale M — so λ captures exactly the part of the future the past lets us anticipate.

2.2 The Poisson process and why deterministic Λ matters

Definition 2.1 (Poisson process). N is an inhomogeneous Poisson process with mean measure Λ if for disjoint bounded sets $A_1, \dots, A_k$ the counts $N(A_i)$ are independent with $N(A_i) \sim \text{Poisson}(\Lambda(A_i))$.

Two consequences are used constantly: a Poisson process has *independent increments*, and $\mathbb{E}[N(t)] = \Lambda(t)$. The deep fact we rely on is the converse.

Theorem 2.2 (Watanabe characterisation). *Let N be a simple point process with continuous compensator Λ . Then N is an inhomogeneous Poisson process with mean measure Λ if and only if Λ is deterministic.*

Intuition. Dependence between increments can only come from the intensity *reacting to history*. If the intensity — hence the compensator — is fixed in advance, there is no channel through which the past can speak to the future, so the process must be Poisson. **This single fact is why the MBPP works:** replacing the random Hawkes intensity λ by its deterministic mean ξ severs the feedback loop and gives us back independent increments (and therefore a product likelihood, §4).

2.3 The likelihood of a point process

Theorem 2.3 (Jacod formula). *A simple point process with $(\mathbb{P}, \mathcal{H}_t)$ -intensity λ has, on $[0, T]$, the log-likelihood (relative to a unit-rate Poisson reference, up to a constant)*

$$\ell(\theta) = \sum_{t_i \leq T} \log \lambda(t_i; \theta) - \int_0^T \lambda(s; \theta) \, ds. \quad (4)$$

Intuition. The likelihood rewards forecasting high intensity exactly where events landed ($\sum \log \lambda(t_i)$) and penalises total forecast mass ($\int \lambda$): “put probability on the bumps, don’t waste it elsewhere.” It is the point-process analogue of $\prod_i p(x_i)$, and it manifestly needs the times t_i — the obstruction in the interval-censored setting.

2.4 The Hawkes process and why it is *not* Poisson

Model 2.4 (Hawkes process). The Hawkes process has stochastic intensity

$$\lambda(t) = s(t) + \int_0^{t-} \phi(t-u) \, dN(u) = s(t) + \sum_{t_i < t} \phi(t-t_i), \quad (5)$$

with exogenous (immigrant) rate s and triggering kernel ϕ . The *exponential kernel* is $\phi(t) = \kappa \theta e^{-\theta t} \mathbb{1}[t > 0]$, for which $\int_0^\infty \phi = \kappa$ is the *branching ratio* and θ the decay rate.

Because λ depends on the realised history, the compensator $\Lambda(t) = \int_0^t \lambda$ is *random*; by Theorem 2.2 the Hawkes process is therefore *not* Poisson and has dependent increments.

2.5 The cluster representation and the mean-intensity equation

Theorem 2.5 (Hawkes–Oakes cluster representation). *The Hawkes process of (5) is distributionally equal to the following branching construction: (i) immigrants arrive as a Poisson process of intensity s ; (ii) each event at time u independently produces direct offspring as a Poisson process of intensity $\phi(\cdot - u)$ on (u, ∞) ; (iii) the process is the superposition of all immigrants and descendants.*

Intuition. Self-excitation made literal: every event founds a “family,” and the whole process is a *forest of independent family trees* rooted at immigrants. The mean number of direct children per event is $\int \phi = \kappa$, so a family is a Galton–Watson tree with mean offspring κ — finite on average iff $\kappa < 1$ (*subcritical*), with expected total size $1/(1 - \kappa)$.

Averaging the cluster representation gives the equation at the heart of everything.

Proposition 2.6 (Mean-intensity / MBPP equation). *Under the standing assumptions of §3, the expected intensity $\xi(t) = \mathbb{E}[\lambda(t)]$ satisfies*

$$\xi(t) = s(t) + \int_0^t \phi(t-u) \xi(u) \, du. \quad (6)$$

Proof. Take $\mathbb{E}[\cdot]$ in (5). By Tonelli (non-negative integrands) the expectation passes inside, and using the tower property $\mathbb{E}[dN(u)] = \mathbb{E}[\mathbb{E}[dN(u) \mid \mathcal{H}_{u-}]] = \mathbb{E}[\lambda(u)] du = \xi(u) du$. $\square$

Intuition. Average the *stochastic* feedback over all realisations and it becomes *deterministic* feedback: “on average, today’s rate is the baseline plus the kernel-weighted echo of the average past.” Equation (6) is the object we will solve again and again. It is a *Volterra integral equation of the second kind* (Appendix A); when — as here — the kernel depends only on $t - u$, it is moreover a *convolution*, which is what makes the next section entirely closed-form.

Remark (Why interval censoring is hard for Hawkes directly). Decomposing a bucket count over offspring generations produces nested *Borel* convolutions with no finite closed form at finite T , and Monte Carlo over the generation structure scales poorly. This is the formal sense in which “the Hawkes process cannot be fit interval-censored” — and the reason we detour through the MBPP.

3 Standing assumptions for our model

We collect, once, the precise regularity conditions under which every result below holds. They are mild and satisfied by every model in the companion package (exponential kernels; power-law kernels with positive offset; non-negative, locally integrable baselines). Each later result states any *extra* hypothesis explicitly.

- (A1) **Simplicity and non-explosion.** N is a simple temporal point process on $[0, \infty)$: no two events coincide ($\mathbb{P}$ -a.s.), and $N(t) < \infty$ a.s. for every finite t .
- (A2) **Stochastic intensity.** N admits an $(\mathcal{H}_t)$ -predictable conditional intensity $\lambda \geq 0$ with respect to its internal history $\mathcal{H}_t = \sigma\{N(s) : s \leq t\}$; equivalently the compensator $\Lambda(t) = \int_0^t \lambda$ is absolutely continuous, with $\int_0^T \lambda < \infty$ a.s.
- (A3) **Kernel.** $\phi : \mathbb{R} \rightarrow [0, \infty)$ is measurable, causal ($\phi(u) = 0$ for $u < 0$), and locally bounded, with branching ratio $n^* := \int_0^\infty \phi = \kappa < \infty$. *Subcritical* means $\kappa < 1$.
- (A4) **Baseline.** $s : [0, \infty) \rightarrow [0, \infty)$ is measurable and locally integrable. Where a Dirac-comb (multi-impulse) baseline appears it is a non-negative measure and convolutions with it are read distributionally.
- (A5) **Finite mean intensity.** Wherever $\mathbb{E}[\lambda]$ is used, $\sup_{t \leq T} \mathbb{E}[\lambda(t)] < \infty$ — automatic under (A3)-subcritical with locally integrable s .
- (A6) **Smooth, identifiable parameterisation.** For estimation, θ ranges over an open $\Theta \subseteq \mathbb{R}^p$; the map $\theta \mapsto \lambda(t; \theta)$ is twice continuously differentiable with $\lambda(t; \theta) > 0$ on the data, and the model is identifiable on Θ .
- (A7) **Multivariate.** In dimension M : $N = (N_1, \dots, N_M)$, intensity $\lambda \in \mathbb{R}_+^M$, kernel *matrix* $\Phi = (\phi_{mj})$, and *branching matrix* $G = (\int \phi_{mj})$. *Stable* means the spectral radius $\rho(G) < 1$.

Intuition. (A1)–(A2) say the process is well-behaved enough to have an instantaneous rate; (A3)–(A4) say the two ingredients ϕ and s are honest non-negative functions with finite total excitation; (A5) keeps means finite; (A6) makes maximum likelihood meaningful; (A7) lifts everything to M interacting components, where the single number $\kappa < 1$ is replaced by the matrix condition $\rho(G) < 1$. Whenever a later theorem needs more — boundedness of ϕ , subcriticality, existence of a transform — it is flagged at that result.

Remark (Two parameterisations of the exponential kernel). The package’s event-time code writes $\phi(t) = \alpha e^{-\beta t}$, while the MBPP theory uses $\phi(t) = \kappa \theta e^{-\theta t}$. They coincide with $\alpha = \kappa \theta$, $\beta = \theta$, i.e. $\kappa = \alpha/\beta$ is the branching ratio. We use (κ, θ) throughout the theory.

4 The MBPP without covariates

Here ϕ is a fixed convolution kernel. The MBPP equation (6) is a *convolution* Volterra equation of the second kind, hence linear and time-invariant (LTI), and everything is closed-form. This is the workhorse the rest of the paper perturbs.

4.1 Existence, uniqueness, and the resolvent

Write the Volterra operator $(\mathcal{V}\xi)(t) = \int_0^t \phi(t-u)\xi(u) du$, so (6) reads $(\mathcal{I} - \mathcal{V})\xi = s$.

Theorem 4.1 (Well-posedness; see Appendix A). *If ϕ is bounded on $[0, T]$ (true for the exponential kernel and for power-law with positive offset) and $s \in C[0, T]$, then (6) has a unique continuous solution on $[0, T]$, given by the Neumann series $\xi = \sum_{n \geq 0} \mathcal{V}^n s$.*

Intuition. On a finite horizon, causal feedback can only compound so fast: the n -th echo $\mathcal{V}^n$ carries a $t^n/n!$ factor that crushes it, so the loop $\mathcal{I} - \mathcal{V}$ is *always* invertible there, whatever the gain κ . “No runaway in finite time.” Subcriticality ($\kappa < 1$) is needed only to extend this to $t = \infty$.

Separating the $n = 0$ term, $\xi = s + h * s$ with the *resolvent kernel* $h = \sum_{n \geq 1} \phi^{*n}$ (ϕ^{*n} the n -fold convolution). The companion impulse response is $E = \delta + h$, so $\xi = E * s$ (Appendix A, B).

Proposition 4.2 (Global resolvent). *If $\kappa = \int_0^\infty \phi < 1$ then $h \in L^1(0, \infty)$ with $\|h\|_1 = \kappa/(1 - \kappa)$, and $\xi = s + h * s$ holds on $[0, \infty)$.*

Intuition. The resolvent sums the echo over *all* offspring generations; it is globally integrable exactly when each generation shrinks ($\kappa < 1$), and its total mass $\kappa/(1 - \kappa)$ is the expected number of offspring per event. The resolvent *is* the renewal density of the branching tree.

4.2 The exponential kernel closes everything

For $\phi(t) = \kappa \theta e^{-\theta t}$ one finds (Appendix C)

$$h(t) = \sum_{n \geq 1} \phi^{*n}(t) = \kappa \theta e^{(\kappa-1)\theta t}, \quad t > 0, \quad (7)$$

a single exponential (the series telescopes because one timescale stays one timescale). The compensator obeys the *same* equation with s replaced by its integral $S(t) = \int_0^t s$:

$$\Xi(t) = S(t) + \int_0^t \phi(t-u) \Xi(u) du = S + h * S, \quad \Xi(x, y] = \Xi(y) - \Xi(x). \quad (8)$$

Example (A single immigrant). For $s = \delta(\cdot - a)$ (one immigrant at a), with $c := (\kappa - 1)\theta < 0$,

$$\xi(t) = \delta(t - a) + \kappa \theta e^{c(t-a)} \mathbb{1}_{[t > a]}, \quad \Xi^{\text{endo}}(t) = \frac{\kappa}{1 - \kappa} (1 - e^{c(t-a)}) \mathbb{1}_{[t > a]}.$$

As $t \rightarrow \infty$, $\Xi^{\text{endo}} \rightarrow \kappa/(1 - \kappa)$: the mean offspring count of one immigrant. The lone event leaves a *decaying wake* of expected offspring whose total mass is the branching sum.

4.3 Multivariate MBPP

With M interacting components, $\xi \in \mathbb{R}^M$ solves the matrix convolution Volterra equation

$$\xi(t) = s(t) + \int_0^t \Phi(t-u) \xi(u) du, \quad \Phi = (\phi_{mj}), \quad (9)$$

where ϕ_{mj} is how component j excites component m . In the transform domain (Appendix B) this is $\hat{\xi}(\omega) = (I - \hat{\Phi}(\omega))^{-1} \hat{s}(\omega)$.

Proposition 4.3 (Stationarity and spectral radius). *Under a constant baseline $s \equiv \bar{\mu}$ and stability $\rho(G) < 1$ (where $G_{mj} = \int \phi_{mj} = \hat{\Phi}(0)_{mj}$ is the branching matrix), the stationary mean intensity is*

$$\bar{\xi} = (I - G)^{-1} \bar{\mu} = \sum_{n \geq 0} G^n \bar{\mu}. \quad (10)$$

Proof. Stationarity in (9) gives $\bar{\xi} = \bar{\mu} + G\bar{\xi}$; the Neumann series converges iff $\rho(G) < 1$. $\square$

Intuition. The stationary intensity is the baseline amplified by the network's *total* feedback $(I - G)^{-1} = I + G + G^2 + \dots$: direct excitation, plus two-hop, plus three-hop, The condition $\rho(G) < 1$ is exactly what makes those multi-hop echoes sum to something finite.

For exponential kernels the matrix equation is realised, with no transform, as a finite *linear ODE* in M^2 states $y_{mj} = (a_{mj} e^{-b_{mj} \cdot}) * \xi$ (Appendix C):

$$y'_{mj} = a_{mj} \xi_j - b_{mj} y_{mj}, \quad \xi_m = s_m + \sum_j y_{mj}, \quad (11)$$

with $a_{mj} = \kappa_{mj} \theta_{mj}$, $b_{mj} = \theta_{mj}$. This is the workhorse solver `solve_mbpp_ode_multivariate`; the numerical check $\bar{\xi} = (I - G)^{-1} \bar{\mu}$ holds to machine precision.

4.4 Fitting on interval-censored counts (IC-LL)

Because the MBPP is Poisson (Theorem 2.2), the bucket counts over an observation grid $0 = o_0 < \dots < o_m = T$ are *independent* Poisson($\Xi(o_{i-1}, o_i]$). The negative log-likelihood is therefore a simple sum.

Proposition 4.4 (Interval-censored log-likelihood). *For observed counts $C(o_{i-1}, o_i]$,*

$$\mathcal{L}_{\text{IC}}(\theta) = \sum_{i=1}^m \Xi(o_{i-1}, o_i; \theta] - \sum_{i=1}^m C(o_{i-1}, o_i] \log \Xi(o_{i-1}, o_i; \theta], \quad (12)$$

up to a constant in θ . Minimising it over the kernel parameters recovers the Hawkes parameters through the MBPP correspondence.

Intuition. Independent Poisson buckets $\Rightarrow$ the joint probability is a product $\Rightarrow$ the log-likelihood is a sum of per-bucket Poisson terms in the interval compensators Ξ_i . It compares a *count* C_i to an *expected count* Ξ_i (both extensive), as it should. (Choosing a Gaussian noise model instead of Poisson turns the loss into least squares; see Appendix D for the Bregman/exponential-family picture and the identifiability of κ versus θ .)

Remark (What is robustly estimable). The branching ratio κ (the gain) is well identified from counts; the timescale θ is weakly identified, because a bucket integrates away the within-bucket timing that carries θ (Appendix D). Treat κ as trustworthy and report θ with wide intervals.

5 Adding covariates to the baseline

The first and easiest place to let data drive the model is the *baseline* (immigrant) rate. Let observed covariates $X(t) \in \mathbb{R}^p$ enter through a log-linear link,

$$s(t) = \exp(\gamma_0 + \gamma^\top X(t)), \quad (13)$$

which keeps $s \geq 0$ for all γ .

5.1 Why this stays easy

The MBPP equation (6) is *linear in the forcing* s and its kernel is unchanged: (13) only changes *what* we feed in, not the operator $\mathcal{I} - \mathcal{V}$. So the entire LTI solution of §4 carries over verbatim, $\xi = E * s$ and $\Xi = E * S$ with $S = \int_0^t s$.

Proposition 5.1 (Piecewise-constant covariates remain closed-form). *If $X(t)$ is piecewise-constant, then $s(t) = \exp(\gamma_0 + \gamma^\top X(t))$ is piecewise-constant, with rate $\exp(\gamma_0 + \gamma^\top X_k)$ on the k -th covariate interval. The MBPP intensity and compensator are then the same closed-form piecewise-constant responses as in §4; only the per-interval rates are reparameterised.*

Intuition. A covariate that is constant on each bucket simply re-labels the height of the baseline on that bucket. The self-excitation machinery does not know or care that the height came from a regression rather than a free parameter — so nothing about the solution method changes, and the closed forms of §4 apply unchanged.

5.2 Fitting the covariate effects

We now optimise (12) over (γ_0, γ) *jointly* with the kernel parameters (κ, θ) . The compensator Ξ is assembled from the log-linear baseline exactly as before, so the only change from §4.4 is a larger but still smooth parameter vector. This is the interval-censored analogue of the event-time covariate model, implemented as `fit_mbpp_ic_covariates`.

Remark (Tabular data). A time-bucketed table — one row per observation interval, with covariate columns X and an event-count column C — maps directly onto the fit: the row endpoints are the observation grid, C is the count vector, and the covariate columns are X . Categorical covariates are one-hot encoded first.

5.3 What is and is not identified

Empirically and theoretically (Appendix D), the covariate *effects* γ are recovered tightly, because they are read off from how counts change *across* buckets as X varies; the absolute split between the baseline level γ_0 and the self-excitation κ is more weakly identified, sharing the κ - θ ridge. A covariate that does not vary across buckets carries no information about its coefficient: identification of γ requires covariate *contrast*.

Intuition. The model can explain a given level of activity either as “a high baseline” or “a modest baseline strongly amplified.” Counts alone struggle to separate the two. But *differences* in activity that track *differences* in a covariate pin that covariate’s effect cleanly — which is why γ is the trustworthy output here.

Crucially, in (13) the kernel ϕ is still a fixed convolution kernel: covariates have entered only the *forcing*. The next section asks what happens when they enter the *kernel* — and that is where the convolution structure, and with it the LTI shortcuts, finally break.

6 Excitation covariates: the Volterra system that arises

We now let covariates modulate the *excitation itself* — the strength with which one event triggers others. This is the genuinely harder case, and the whole point of developing the Volterra viewpoint: the governing equation stops being a convolution.

6.1 The model

Model 6.1 (Covariate-modulated excitation). Let observed covariates $Z(t) \in \mathbb{R}^q$ scale the triggering strength through a log-linear link,

$$\alpha_{mj}(t) = \alpha_{mj}^0 \exp(\delta_{mj}^\top Z(t)), \quad \phi_{mj}(t, u) = \alpha_{mj}(t) g_{mj}(t - u), \quad (14)$$

where $g_{mj} \geq 0$ is a fixed shape (e.g. $g_{mj}(\tau) = \theta_{mj} e^{-\theta_{mj}\tau}$) and $\alpha_{mj}^0 \geq 0$ is the covariate-free base strength.

The modulation may be read at the *receiving* time t (as written) or at the *parent* time u (replace $Z(t)$ by $Z(u)$); both are handled below and the implementation uses the former.

6.2 The mean-intensity equation is now a general Volterra equation

Averaging the Hawkes intensity exactly as in Proposition 2.6, but with the time-dependent kernel (14), gives

Proposition 6.2 (Non-convolution MBPP equation). *Under (A1)–(A7), the expected intensity solves the system of linear Volterra equations of the second kind*

$$\xi_m(t) = s_m(t) + \sum_{j=1}^M \int_0^t K_{mj}(t, u) \xi_j(u) du, \quad K_{mj}(t, u) = \alpha_{mj}^0 e^{\delta_{mj}^\top Z(t)} g_{mj}(t - u). \quad (15)$$

The kernel $K_{mj}(t, u)$ depends on t and u separately, not only on $t - u$.

Intuition. In §4–5 the kernel was a function of the elapsed time $t - u$ alone — the system reacted to a stimulus the same way regardless of *when* it arrived. Now the response strength rides on $Z(t)$, so *when* matters: the same past event echoes more loudly in a high- Z period than in a low- Z period. Mathematically the kernel is no longer a convolution, and **this is exactly what costs us the LTI toolkit**.

6.3 What survives, and what is lost

Well-posedness survives.

Equation (15) is a standard linear Volterra system. By the general theory (Appendix A, which does *not* use convolution structure), for any locally bounded kernel and continuous s it has a unique solution on every finite horizon, given by the Neumann series $\xi = \sum_{n \geq 0} \mathcal{K}^n s$ with $(\mathcal{K}\xi)_m(t) = \sum_j \int_0^t K_{mj}(t, u) \xi_j(u) du$.

Time-invariance is lost.

Because $K(t, u) \neq K(t - u)$, the convolution theorem no longer applies: there is *no* Laplace/Fourier solution, *no* constant transfer function $R(\omega)$, and the spectral (Fourier-neural-operator) shortcut of the convolution case is no longer exact. The resolvent is now a genuine two-variable kernel $R(t, u)$ solving $R = K + K \circ R$ (Appendix A), not a single function of $t - u$.

6.4 Exponential shape $\Rightarrow$ a finite linear time-varying ODE

All is not lost computationally. The key is that an exponential g makes the kernel *separable* (degenerate): with $g_{mj}(\tau) = e^{-b_{mj}\tau}$,

$$K_{mj}(t, u) = \underbrace{\alpha_{mj}^0 e^{\delta_{mj}^\top Z(t)} e^{-b_{mj}t}}_{a_{mj}(t)} \underbrace{e^{b_{mj}u}}_{(u \text{ only})},$$

a rank-one product of a t -function and a u -function. A separable Volterra kernel collapses the integral equation onto finitely many ODEs (Appendix A). Concretely, introducing the filtered histories $y_{mj}(t) = \int_0^t e^{-b_{mj}(t-u)} \xi_j(u) du$:

Theorem 6.3 (LTV state-space reduction). *For exponential shapes, the non-convolution MBPP equation (15) (receiving-time modulation) is equivalent to the finite linear time-varying ODE system*

$$y'_{mj}(t) = \xi_j(t) - b_{mj} y_{mj}(t), \quad \xi_m(t) = s_m(t) + \sum_j \alpha_{mj}^0 e^{\delta_{mj}^\top Z(t)} y_{mj}(t), \quad (16)$$

with $y_{mj}(0) = 0$. The coefficients are constant except for the modulation $e^{\delta^\top Z(t)}$, which makes the system time-varying rather than time-invariant.

Intuition. The history of each component is still summarised by a finite “running, decaying memory” y_{mj} — exactly the state of the convolution case (11). The *only* change is that the gain with which that memory feeds back, $\alpha_{mj}^0 e^{\delta^\top Z(t)}$, now breathes in and out with the covariate. So we keep a finite-dimensional, integrable system; we merely lose the constant-coefficient (and hence transform-solvable) structure. “Same state, time-varying gain.”

The system (16) is integrated by Runge–Kutta exactly like the convolution case; the package provides it as `solve_mbpp_ltv`. Two checks confirm correctness: with $Z \equiv 0$ (no modulation) it reproduces the LTI multivariate solver to machine precision ($\sim 10^{-16}$); and with a covariate that, say, *doubles* the excitation over a window, the intensity rises sharply during the window and *relaxes back* to the unmodulated stationary level afterwards — the signature of a stable forced system.

Remark (Power-law shapes: direct Volterra quadrature). If g is *not* a finite sum of exponentials (e.g. power-law), the kernel is *not* separable and no finite ODE exists. One then solves (15) directly by discretising the Volterra integral on the observation grid — a triangular (lower-block-triangular) linear system solved by forward substitution in $O(N^2 M^2)$ (the Nyström method, Appendix A). One may also approximate g by a sum of exponentials (Prony) to restore a finite LTV ODE to any desired accuracy.

6.5 Fitting with excitation covariates

The MBPP is *still* a Poisson process with a deterministic (now harder-to-compute) intensity, so the interval-censored likelihood (12) still applies verbatim: assemble $\Xi(o_{i-1}, o_i]$ by integrating the LTV ODE (16) (or by quadrature), and minimise $\mathcal{L}_{IC}$ over $(\alpha^0, \theta, \delta, \gamma)$. Three things change relative to the convolution case:

1. **More parameters.** Each excitation-covariate coefficient δ_{mj} adds dimensions; with p covariates and an $M \times M$ network that is up to pM^2 extra parameters. An ℓ_1 penalty or low-rank/shared- δ structure is advisable.
2. **Non-convexity.** The δ enter through $e^{\delta^\top Z}$ *inside* the intensity, so the loss is non-convex (unlike the baseline-only case, which is well-behaved). Use multiple restarts.

3. **Identification needs interaction.** A δ_{mj} is identified only if component j actually fires while Z varies; without that co-occurrence the excitation modulation leaves no fingerprint in the counts.

Intuition. We have traded a closed-form, convex, transform-solvable problem for a finite-dimensional but *time-varying* and *non-convex* one. The physics is appealing — “contagion that is more contagious in some regimes than others” — and entirely doable; one simply pays for it in computation (integrate an ODE instead of evaluating a formula) and in statistics (more parameters, a rougher landscape, a stronger need for covariate variation). That trade is the precise content of “excitation covariates are harder.”

6.6 Summary of the three regimes

	§4/5 no/baseline cov.	§6 excitation covariates	
	convolution	exp. shape	general shape
MBPP equation	convolution Volterra	general Volterra (non-convolution)	
Time-invariant?	yes (LTI)	no	no
Laplace/Fourier solution	yes	no	no
Reduces to finite ODE?	yes (constant coef.)	yes (<i>time-varying</i> coef.)	no
How we compute ξ, Ξ	closed form	integrate LTV ODE	Volterra quadrature
Fitting loss	convex(ish)	non-convex	non-convex

Reading the table left to right is reading this document: each added layer of realism removes a layer of mathematical convenience, and the Volterra formulation is what lets us track exactly which convenience is lost and what replaces it.

A Volterra integral equations: a primer

This appendix is self-contained and assumes no prior exposure to integral equations. Everything the main text needs about Volterra equations is here.

A.1 What is an integral equation, and why “Volterra”?

An *integral equation* has the unknown function under an integral sign, just as a *differential equation* has it under a derivative. The kind we meet is the *linear Volterra equation of the second kind*:

$$\xi(t) = s(t) + \int_0^t K(t, u) \xi(u) du, \quad t \in [0, T]. \quad (17)$$

Here s (the “forcing” or “free term”) and K (the “kernel”) are given, and ξ is unknown. “Second kind” means the unknown also appears *outside* the integral (the lone $\xi(t)$ on the left); a *first-kind* equation $s(t) = \int_0^t K(t, u) \xi(u) du$ has the unknown only inside and is much more delicate (it is a deconvolution and is typically ill-posed).

The name *Volterra* refers to the upper limit being the running time t : the integral only looks at the *past* $u \leq t$. Contrast a *Fredholm* equation, whose integral runs over a fixed full interval $\int_0^T$. The Volterra (causal) structure is what makes (17) so well-behaved: it is, in a precise sense, “lower-triangular in time.”

Intuition. Think of (17) as a feedback rule unrolling in time: the value $\xi(t)$ “now” equals an external input $s(t)$ *plus* a weighted echo of everything $\xi(u)$ that happened before, with weight $K(t, u)$. Because the echo only reaches back into the past, you can in principle sweep forward and build ξ left to right — nothing in the future is needed to determine the present. That causal, triangular character is the source of all the good news below.

A.2 The operator form and the Neumann (resolvent) series

Define the *Volterra operator*

$$(\mathcal{K}\xi)(t) := \int_0^t K(t, u) \xi(u) du, \quad (18)$$

so (17) is $\xi = s + \mathcal{K}\xi$, i.e. $(\mathcal{I} - \mathcal{K})\xi = s$. Formally inverting the operator as a geometric series,

$$\xi = (\mathcal{I} - \mathcal{K})^{-1}s = \sum_{n=0}^{\infty} \mathcal{K}^n s = s + \mathcal{K}s + \mathcal{K}^2 s + \dots \quad (19)$$

This is the *Neumann series*. The operator $\mathcal{K}^n$ has its own kernel, the *iterated kernel* $K_n(t, u)$ (with $K_1 = K$ and $K_{n+1}(t, u) = \int_u^t K(t, r) K_n(r, u) dr$), and the sum $R(t, u) = \sum_{n \geq 1} K_n(t, u)$ is the *resolvent kernel*. With it,

$$\xi(t) = s(t) + \int_0^t R(t, u) s(u) du, \quad (20)$$

so solving the equation amounts to computing R *once* and then a single integral against any forcing s . R satisfies the resolvent equations $R = K + K \circ R = K + R \circ K$, where $\circ$ is the Volterra composition above.

Intuition. $\mathcal{K}s$ is the first-generation echo of the input, $\mathcal{K}^2 s$ the echo of the echo, and so on; (19) simply sums all generations. The resolvent R packages every generation into one kernel — it is the “total-feedback” Green’s function of the problem. (In the Hawkes setting R is literally the renewal density of the branching tree, §4.)

A.3 Existence and uniqueness: why Volterra is so forgiving

Theorem A.1 (Existence and uniqueness on a finite horizon). *Suppose K is bounded, $|K(t, u)| \leq \bar{K}$ on the triangle $0 \leq u \leq t \leq T$, and $s \in C[0, T]$. Then (17) has a unique continuous solution on $[0, T]$, namely the (uniformly convergent) Neumann series (19).*

Proof. By induction the iterated operator satisfies $|(\mathcal{K}^n \xi)(t)| \leq \bar{K}^n \|\xi\|_{\infty} t^n / n!$: indeed $|(\mathcal{K}\xi)(t)| \leq \bar{K} \|\xi\|_{\infty} t$, and $|(\mathcal{K}^{n+1} \xi)(t)| \leq \int_0^t \bar{K} |(\mathcal{K}^n \xi)(r)| dr \leq \bar{K}^{n+1} \|\xi\|_{\infty} \int_0^t r^n / n! dr = \bar{K}^{n+1} \|\xi\|_{\infty} t^{n+1} / (n+1)!$. Hence $\|\mathcal{K}^n\|_{C[0, T]} \leq (\bar{K}T)^n / n! \rightarrow 0$, the series $\sum_n \mathcal{K}^n$ converges in operator norm, $(\mathcal{I} - \mathcal{K})$ is boundedly invertible, and the fixed point (19) is unique. $\square$

Intuition. The decisive estimate is the factorial $\bar{K}^n t^n / n!$. Each time the echo passes through the kernel it must integrate over a *shrinking* triangle of “time still available,” and n such nested integrations contribute a $1/n!$ that overwhelms any fixed gain $\bar{K}^n$. So on a finite horizon the loop is *always* invertible — there is no smallness condition on the kernel at all. This is the single most important structural fact, and it is why the MBPP equation is solvable even when the excitation is covariate-modulated and time-varying (§6): nothing in the proof used convolution structure.

A smallness condition *does* appear if one wants the resolvent globally on $[0, \infty)$: for a convolution kernel $K(t, u) = \phi(t - u)$, Young’s inequality gives $\|\phi^{*n}\|_1 \leq \|\phi\|_1^n$, so $R = \sum_n \phi^{*n} \in L^1(0, \infty)$ iff $\|\phi\|_1 = \kappa < 1$ — exactly subcriticality.

A.4 Three solution strategies, by kernel type

The structure of K dictates the best method. The main text uses all three.

(i) **Convolution kernel** $K(t, u) = \phi(t - u) \Rightarrow$ **transforms / LTI**. Then $\mathcal{K}\xi = \phi * \xi$ and the system is time-invariant. Laplace or Fourier turn the convolution into a product (Appendix B): $\hat{\xi} = \hat{s}/(1 - \hat{\phi})$. The resolvent is itself a convolution, $R(t, u) = h(t - u)$ with $h = \sum_n \phi^{*n}$, and for exponential ϕ it is a single exponential. *This is §4–5.*

(ii) **Separable (degenerate) kernel** $K(t, u) = \sum_{r=1}^{\rho} a_r(t) b_r(u) \Rightarrow$ **finite ODE**. A finite-rank kernel collapses the integral equation onto ρ ordinary differential equations. Indeed, defining the states $y_r(t) = \int_0^t b_r(u) \xi(u) du$, the equation (17) becomes $\xi(t) = s(t) + \sum_r a_r(t) y_r(t)$, and differentiating each state gives the closed ODE system

$$y_r'(t) = b_r(t) \xi(t) = b_r(t) \left(s(t) + \sum_{r'} a_{r'}(t) y_{r'}(t) \right), \quad y_r(0) = 0. \quad (21)$$

If the a_r, b_r are constants (or pure exponentials), (21) has *constant* coefficients — a linear time-invariant ODE; if the a_r carry an extra t -dependence (as the covariate factor $e^{\delta^\top Z(t)}$ does), it is a *time-varying* linear ODE. *An exponential triggering shape is separable of rank one per component pair, which is precisely why §6 reduces to the LTV system (16).*

(iii) **General kernel** $\Rightarrow$ **numerical quadrature (Nyström)**. When K is neither convolution nor separable, discretise the integral on a grid $0 = u_0 < \dots < u_N = T$. Replacing $\int_0^{t_i}$ by a quadrature rule turns (17) into a *lower-triangular* linear system

$$\xi_i = s_i + \sum_{k \leq i} w_{ik} K(t_i, u_k) \xi_k, \quad i = 0, \dots, N, \quad (22)$$

which is solved by forward substitution in $O(N^2)$ (the diagonal term is handled by isolating ξ_i). Convergence is at the order of the quadrature rule. This is the fallback for, e.g., a power-law shape under covariate-modulated excitation.

Intuition. The three strategies are the same idea — invert $\mathcal{I} - \mathcal{K}$ — specialised to how much structure K has. Maximum structure (convolution): a formula. Medium structure (finite rank): a handful of ODEs. No structure: march forward in time filling in ξ one step at a time, which the triangular (causal) form always permits.

Remark (First-kind equations and why we avoid them). The compensator’s *inverse* problem — recovering ϕ from observed mean behaviour — is a first-kind (deconvolution) Volterra problem and is ill-posed (small data errors blow up). The main text never inverts in this direction for estimation; it always poses a well-posed second-kind forward solve inside a likelihood, which is far more stable.

B Laplace and Fourier methods for convolution kernels

The transform methods of this appendix apply *only* to the convolution case $K(t, u) = \phi(t - u)$ (no/baseline covariates). Their failure for non-convolution kernels is exactly what makes the excitation-covariate case (§6) hard.

B.1 The transforms and the convolution theorem

The (one-sided) Laplace transform of f is $\mathcal{L}\{f\}(z) = \int_0^\infty e^{-zt} f(t) dt$, and the Fourier transform is $\mathcal{F}\{f\}(\omega) = \hat{f}(\omega) = \int_{-\infty}^\infty e^{-i\omega t} f(t) dt$. The one property we need is that both turn convolution into multiplication:

$$\mathcal{L}\{\phi * \xi\} = \mathcal{L}\{\phi\} \mathcal{L}\{\xi\}, \quad \widehat{\phi * \xi} = \hat{\phi} \hat{\xi}. \quad (23)$$

Intuition. A convolution mixes every past value of ξ into the present through a *shift-invariant* weight. The transforms re-express ξ in terms of pure oscillations $e^{i\omega t}$ (or modes e^{zt}), and a shift-invariant system acts on each such mode merely by scaling it. So in the transform domain the tangled convolution becomes an independent, per-frequency multiplication — the system is “diagonalised.”

B.2 Solving the convolution Volterra equation

Apply $\mathcal{L}$ to $\xi = s + \phi * \xi$: $\mathcal{L}\{\xi\} = \mathcal{L}\{s\} + \mathcal{L}\{\phi\} \mathcal{L}\{\xi\}$, hence

$$\mathcal{L}\{\xi\}(z) = \frac{\mathcal{L}\{s\}(z)}{1 - \mathcal{L}\{\phi\}(z)}, \quad \hat{\xi}(\omega) = R(\omega) \hat{s}(\omega), \quad R(\omega) = \frac{1}{1 - \hat{\phi}(\omega)}. \quad (24)$$

The multiplier R is the *transfer function*; inverting the transform recovers ξ . $R = 1/(1 - \hat{\phi})$ is the frequency-domain face of the geometric series $\sum_n \hat{\phi}^n$ — the resolvent of Appendix A.

Example (Exponential kernel). For $\phi(t) = \kappa\theta e^{-\theta t}$, $\mathcal{L}\{\phi\}(z) = \kappa\theta/(z + \theta)$, so

$$1 - \mathcal{L}\{\phi\} = \frac{z + (1 - \kappa)\theta}{z + \theta}, \quad R(\omega) = \frac{\theta + i\omega}{(1 - \kappa)\theta + i\omega}.$$

This is a one-pole filter; its pole at $-(1 - \kappa)\theta$ is precisely the rate of the ODE in Appendix C, and inverting $\mathcal{L}\{\phi\}/(1 - \mathcal{L}\{\phi\})$ gives the single-exponential resolvent $h(t) = \kappa\theta e^{(\kappa-1)\theta t}$. Time domain, frequency domain, and state space are three faces of one operator.

In the multivariate case the same computation gives $\hat{\xi}(\omega) = (I - \hat{\Phi}(\omega))^{-1} \hat{s}(\omega)$, and at $\omega = 0$ the matrix $\hat{\Phi}(0) = G$ is the branching matrix, recovering the stationary law $\bar{\xi} = (I - G)^{-1} \bar{\mu}$ (Proposition 4.3).

B.3 The spectral (neural-operator) viewpoint, and why it is exact here

Because (24) is a pointwise multiply in frequency, the solution operator $s \mapsto \xi$ is *diagonal* under the Fourier transform. A single linear Fourier-neural-operator layer computes exactly $\xi = \mathcal{F}^{-1}\{R \cdot \mathcal{F}\{s\}\}$, so with $R = 1/(1 - \hat{\phi})$ it represents the MBPP operator with *no* approximation; conversely, *learning* $R(\omega)$ from input/output pairs is nonparametric identification of the operator (and hence of ϕ via $\hat{\phi} = 1 - 1/R$).

B.4 Why transforms fail under excitation covariates

All of the above hinges on $K(t, u) = \phi(t - u)$. When the excitation is covariate-modulated, $K(t, u) = \alpha^0 e^{\delta^\top Z(t)} g(t - u)$ depends on t and u *separately*; the convolution theorem no longer applies, there is no single transfer function, and the operator is *not* diagonalised by Fourier. One must return to the time domain — the LTV ODE or Volterra quadrature of Appendix A. This is the precise mathematical price of §6.

Intuition. Transforms diagonalise *shift-invariant* systems. A covariate that changes the gain over time breaks shift-invariance — the system treats “now” differently from “then” — so no single set of frequency weights can describe it. The cure is to stop trying to diagonalise and instead integrate the dynamics forward in time.

C ODE and state-space methods

The bridge between integral equations and differential equations: when the kernel is built from exponentials, the Volterra equation *is* an ODE in disguise.

C.1 The exponential kernel gives a first-order ODE

Take the convolution MBPP equation $\xi = s + \phi * \xi$ with $\phi(t) = \kappa\theta e^{-\theta t}$, and split off the endogenous part $y := \xi - s = \phi * \xi$. Differentiating $y(t) = \kappa\theta \int_0^t e^{-\theta(t-u)} \xi(u) du$ (Leibniz rule),

$$y'(t) = \kappa\theta \xi(t) - \theta y(t) = (\kappa - 1)\theta y(t) + \kappa\theta s(t), \quad y(0) = 0, \quad \xi = s + y. \quad (25)$$

So ξ obeys a *first-order linear ODE forced by the baseline s* , with a *stable* pole at $(\kappa - 1)\theta < 0$. Solving (25) with the integrating factor reproduces the single-exponential resolvent $h(t) = \kappa\theta e^{(\kappa - 1)\theta t}$.

Intuition. A single memory timescale is a single pole, i.e. a first-order filter, i.e. a first-order ODE. One number κ sets the feedback gain and one number θ sets the relaxation rate $(1 - \kappa)\theta$: push the system and it returns to baseline at that rate.

C.2 Rational Laplace transform $\Leftrightarrow$ finite ODE

Theorem C.1 (Realisation). *The convolution Volterra equation $\xi = s + \phi * \xi$ is equivalent to a finite-order linear constant-coefficient ODE (forced by s and its derivatives) if and only if ϕ has a strictly proper rational Laplace transform — equivalently, iff ϕ is a finite exponential-polynomial $\sum_q p_q(t)e^{-b_q t}$.*

Proof. ($\Leftarrow$) If $\mathcal{L}\{\phi\} = P/Q$ with $\deg P < \deg Q =: d$, then from (24) $\mathcal{L}\{\xi\} = \frac{Q}{Q-P}\mathcal{L}\{s\}$, so $(Q - P)(z)\mathcal{L}\{\xi\} = Q(z)\mathcal{L}\{s\}$. As $Q - P$ has degree d , reading $z^k \leftrightarrow d^k/dt^k$ yields the order- d ODE $(Q - P)(\frac{d}{dt})\xi = Q(\frac{d}{dt})s$. ($\Rightarrow$) Conversely a finite ODE makes the transfer function $Q/(Q - P)$ rational, hence $\mathcal{L}\{\phi\} = 1 - (Q - P)/Q = P/Q$ rational; and a function has rational Laplace transform iff it is an exponential-polynomial (partial fractions and inverse Laplace). $\square$

Intuition. A memory built from finitely many decaying modes can be tracked by finitely many state variables — “rational transfer function $\Leftrightarrow$ finite-dimensional machine” (this is realisation theory). A power-law memory is *not* rational: it carries infinitely many timescales, so no finite ODE exists (one then uses Volterra quadrature, Appendix A).

C.3 Sums of exponentials: the state-space realisation

For $\phi(\tau) = \sum_q a_q e^{-b_q \tau}$ the natural states are $y_q = (a_q e^{-b_q \cdot}) * \xi$, giving the constant-coefficient system

$$y'_q = a_q \xi - b_q y_q, \quad \xi = s + \sum_q y_q, \quad (26)$$

i.e. $\dot{u} = Au + a s$, $\xi = s + \mathbf{1}^\top u$ with $A = a\mathbf{1}^\top - \text{diag}(b)$. The multivariate version (kernel matrix) carries an index pair (m, j) and gives the M^2 -state system (11), implemented as `solve_mbpp_ode_multivariate`. Since sums of exponentials are dense, this realises (or approximates) essentially any kernel.

C.4 Numerical integration and the time-varying case

These ODEs are integrated by a standard explicit Runge–Kutta (RK4) sweep on the observation grid; the compensator $\Xi = \int_0^t \xi$ is accumulated alongside (an extra trapezoidal state). The constant-coefficient systems are autonomous in their coefficients; the *covariate-modulated* system (16) is identical in form but with the readout gain $\alpha_{mj}^0 e^{\delta_{mj}^\top Z(t)}$ depending on t — a *linear time-varying* (LTV) ODE.

Proposition C.2 (LTV well-posedness). *If the coefficient maps $t \mapsto A(t)$ and $t \mapsto a(t)$ are continuous (here through the continuous covariate path $Z(t)$), the LTV system $\dot{u} = A(t)u + a(t)s(t)$ has a unique solution on $[0, T]$ for every initial condition, given by the time-ordered (Peano–Baker) series; numerically, RK4 converges at its usual order.*

Intuition. A time-varying linear ODE is no harder to *integrate* than a constant one — you just re-read the coefficients at each step. What you lose is the *closed form*: there is no single matrix exponential e^{At} when A depends on t (the $A(t)$ at different times need not commute), which is the differential-equations echo of “no constant transfer function” from Appendix B. So `solve_mbpp_ltv` integrates (16) step by step, and that is exactly how the excitation-covariate model of §6 is computed.

D Estimation, Bregman geometry, and identifiability

D.1 IC-LL is a Bregman divergence

Write the interval counts $\mathbf{C} = (\mathbf{C}_i)$ and compensators $\Xi(\theta) = (\Xi_i(\theta))$. A *Bregman divergence* for a strictly convex generator φ is $B_\varphi(x, y) = \varphi(x) - \varphi(y) - \langle x - y, \nabla \varphi(y) \rangle \geq 0$.

Proposition D.1. *With the generalized-KL generator $\varphi(x) = \sum_i (x_i \log x_i - x_i)$, $\mathcal{L}_{\text{IC}}(\theta) = \text{KL}(\mathbf{C}, \Xi(\theta)) + \Gamma(\mathbf{C})$ with Γ free of θ ; hence $\arg \min_\theta \mathcal{L}_{\text{IC}} = \arg \min_\theta \text{KL}(\mathbf{C}, \Xi(\theta))$. The squared-error generator $\frac{1}{2}\|x\|^2$ instead yields the least-squares loss $\sum_i (\mathbf{C}_i - \Xi_i)^2$.*

Theorem D.2 (Bregman $\leftrightarrow$ exponential family; Banerjee et al.). *For a regular exponential family with cumulant ψ and mean $\mu = \nabla \psi$, the negative log-likelihood equals $B_{\psi^*}(x, \mu)$ up to a constant, where ψ^* is the convex conjugate. Poisson gives $\psi^*(\mu) = \mu \log \mu - \mu$ (so $B = \text{KL}$); Gaussian gives $\psi^*(\mu) = \mu^2/2$ (so $B = \text{squared error}$).*

Intuition. Maximum likelihood in any exponential family is “move the model mean as close as possible to the data, in the geometry set by the family’s variance function.” Choosing the divergence is choosing the noise model: $x \log x \rightarrow \text{Poisson} \rightarrow \text{KL} \rightarrow \text{IC-LL}$; $\frac{1}{2}x^2 \rightarrow \text{Gaussian} \rightarrow \text{least squares}$. Because a Poisson bucket has variance equal to its mean, KL correctly down-weights large-count buckets by $1/\Xi_i$, whereas least squares treats all buckets alike and is dominated by the loud ones — so KL is the more robust loss when counts span orders of magnitude (bursts, near criticality).

D.2 Consistency, information, and goodness of fit

Theorem D.3 (Asymptotics of the MLE; Ogata). *For a stationary, ergodic, subcritical process with true θ_0 in the interior of a compact, identifiable Θ , twice-differentiable log-intensity with integrable dominating envelopes, and finite nonsingular per-time information $\mathbf{i}(\theta_0)$, the MLE satisfies $\hat{\theta}_T \rightarrow \theta_0$ a.s. and $\sqrt{T}(\hat{\theta}_T - \theta_0) \rightarrow \mathcal{N}(0, \mathbf{i}(\theta_0)^{-1})$.*

The information is $\mathcal{I}(\theta) = \mathbb{E} \int_0^T \frac{\partial_\theta \lambda \partial_\theta \lambda^\top}{\lambda} dt$ (event-time) or $\mathcal{I}_{jk}^{\text{IC}} = \sum_i \partial_{\theta_j} \Xi_i \partial_{\theta_k} \Xi_i / \Xi_i$ (interval-censored); standard errors come from inverting the observed information (the Hessian of $-\ell$), and Cramér–Rao says no unbiased estimator beats $\mathcal{I}^{-1}$.

Theorem D.4 (Time-rescaling; goodness of fit). *If Λ is continuous, strictly increasing with $\Lambda(\infty) = \infty$, the rescaled times $\Lambda(t_k)$ form a unit-rate Poisson process; equivalently the rescaled gaps $\Lambda(t_k) - \Lambda(t_{k-1})$ are i.i.d. $\text{Exp}(1)$.*

Intuition. Measure time not in seconds but in *accumulated compensator* — “operational time” that ticks fast when the intensity is high. In that clock the events look like a plain unit-rate Poisson process, giving a model-free residual check (rescaled gaps should be $\text{Exp}(1)$).

D.3 The κ – θ ridge, and the cost of covariates

Proposition D.5 (Weak identifiability of the timescale). *In $\phi = \alpha e^{-\beta t}$, $\alpha = \kappa\theta$, $\beta = \theta$, the (α, β) information matrix is typically ill-conditioned, with its small-eigenvalue direction aligned with curves of constant $\alpha/\beta = \kappa$. Hence κ is well identified while θ is weak, and the more so the coarser the censoring: as the bucket width grows the timing-borne sensitivity $\partial_\theta \Xi_i \rightarrow 0$ while $\partial_\kappa \Xi_i$ stays order one.*

Intuition. The counts pin down “how much” (the gain κ) but not “how fast” (the timescale θ), because a bucket integrates away the within-bucket timing that carries θ . So treat κ and the covariate effects γ as the trustworthy estimands; fix or prior-constrain θ from domain knowledge and report it with wide intervals.

Excitation covariates add non-convexity. Baseline covariates (§5) enter the compensator linearly through a log-link in the *forcing* and keep the problem well-behaved; the covariate effect γ is recovered tightly. Excitation covariates (§6) enter through $e^{\delta^\top Z}$ *inside* the intensity dynamics, so the likelihood is non-convex in δ , with up to pM^2 extra parameters and a real need for covariate–event co-occurrence to identify each δ_{mj} . This is the statistical half of “excitation covariates are harder”; the computational half is that Ξ now comes from integrating an LTV ODE rather than from a formula.

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